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LLM-Guided Visual and Semantic Alignment for Medical Image Classification

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ABSTRACT Vision-language models (VLMs) have shown increasing potential for medical image classification, particularly when prompt learning is used to adapt frozen pretrained encoders to small annotated datasets. Recent methods such as ViP and MDVLA enrich class prompts with LLM-generated symptom descriptions and multi-angle visual features, but they still leave three practical issues unresolved: unfiltered LLM outputs may inject non-imaging or hallucinated descriptions into class prototypes, visual and textual representations are usually computed independently, and the stability of such systems across LLMs and medical knowledge bases is rarely quantified. This paper proposes *LLM-Guided Visual and Semantic Alignment* (LLM-VSA), a frozen-encoder framework that combines filtered LLM-derived visual knowledge with per-view visual features through cross-modal prompt fusion. A Visual Symptom Generator applies redundancy removal, noise/irrelevance filtering, and knowledge-base intersection to retain imaging-relevant descriptions. A Prompt Fusion Module then lets class-specific textual queries attend to per-view visual key vectors before final classification. On Derm7pt, LLM-VSA achieves 0.8237 accuracy and 0.7870 macro-F1, improving over ViP by 1.26 and 1.40 percentage points, respectively ($p < 0.001$, paired t -test; Cohen's $d = 4.16$). The framework also achieves 0.8795 accuracy and 0.8620 macro-F1 on Pneumonia. Additional experiments analyse component contributions, filtering sensitivity, LLM variability, ontology sensitivity, parameter efficiency, and the FLOPs–accuracy trade-off. Code and generation logs are available at <https://github.com/MansoorHayat777/llm-guided-medical-vlp>.

INDEX TERMS CLIP, large language models, medical image classification, multi-angle visual enhancement, prompt learning, vision-language models, visual-semantic alignment

I. INTRODUCTION

MEDICAL image classification often depends on subtle visual cues and limited labelled data. Vision-language models (VLMs) offer a useful starting point because they align images and text in a shared embedding space, but their general-domain pretraining does not directly solve the medical adaptation problem. Methods such as MedCLIP [10] and MedKLIP [12] reduce this gap through clinical text supervision, while prompt learning methods adapt frozen encoders with a small number of trainable parameters. In this direction, ViP [8] uses LLM-generated symptom descriptions to enrich medical prompts, and MDVLA [9] adds multi-angle

visual features to improve orientation robustness.

These methods establish an important direction, but three issues remain. First, LLM-generated descriptions are often inserted into class prompts without explicit filtering. This can introduce redundant, systemic, or non-imaging statements into the class prototype. Second, visual and textual branches are commonly computed independently and compared only through cosine similarity at inference time. This limits the ability of the semantic branch to attend selectively to discriminative visual views. Third, the stability of LLM-guided prompt learning across different LLMs and medical knowledge bases is rarely reported, although this is important for

reproducibility and practical adoption.

This paper proposes *LLM-Guided Visual and Semantic Alignment* (LLM-VSA), a targeted extension of LLM-augmented prompt learning for medical image classification. The framework keeps the pretrained image and text encoders frozen, filters LLM-generated symptom descriptions through a three-rule Visual Symptom Generator (VSG), and fuses the resulting semantic knowledge with orientation-diverse visual features through a Prompt Fusion Module (PFM). The PFM operates on per-view key vectors before view aggregation, so class-specific textual queries can attend to individual visual views rather than a collapsed representation.

The main contributions are as follows.

- We propose a three-rule VSG filtering pipeline that removes redundant, non-imaging, and ontology-inconsistent LLM outputs before they are used as class knowledge.
- We introduce a cross-modal PFM in which learnable textual queries attend to per-view visual key vectors, addressing the independent-branch limitation of earlier prompt-learning methods.
- We report state-of-the-art results on Derm7pt and Pneumonia against prompt-learning and extended VLM baselines, with statistical testing over three backbones and three random seeds.
- We provide controlled ablations for component design, filtering rules, number of views, number of LLM descriptions, PFM variants, parameter efficiency, FLOPs–accuracy trade-off, cross-LLM stability, and ontology sensitivity.

II. RELATED WORK

Medical VLMs aim to transfer the image–text alignment of general models to clinical domains. Earlier deep learning studies established strong image-only baselines in dermatology and chest X-ray analysis [1], [2], [14], while CLIP showed that large-scale image–text pretraining can produce transferable visual representations [3]. Its medical transfer has also been studied through lightweight CLIP adaptation [16]. Subsequent medical VLMs use clinical text more directly: report-supervised learning showed that clinical language can provide richer supervision than image-level labels alone [13], MedCLIP [10] extends contrastive learning to unpaired medical image–text data, and PMC-CLIP [11] and MedKLIP [12] improve domain relevance through large-scale biomedical pretraining. These approaches are effective but usually require large curated corpora or additional pretraining, which can limit their applicability when only small task-specific datasets are available.

Prompt learning offers a lighter adaptation route. CoOp [4] learns continuous context tokens, CoCoOp [5] conditions prompts on image features, KgCoOp [6] adds external textual knowledge, and MaPLe [7] jointly optimizes visual and textual prompts. These methods reduce the number of trainable parameters, but in medical settings they often depend on short

class names or generic phrases that do not encode the visual findings clinicians use for diagnosis.

LLM-augmented prompt learning addresses this semantic gap by generating class-level descriptions. ViP [8] uses LLM-generated symptom prompts to enrich prototypes, and MDVLA [9] adds multi-angle visual representations. Related work has also explored medical foundation models and instruction-tuned assistants [20]–[22], ontology-grounded medical knowledge use [15], hierarchical semantic representations [17], cross-modal interaction modules [18], and multi-view visual modelling [19]. LLM-VSA builds on this line of work but adds two controls that are important for medical use: systematic filtering of LLM-derived descriptions and explicit cross-modal fusion between class semantics and per-view visual features. The framework also evaluates sensitivity to both LLM choice and medical ontology choice, which helps separate method-level effects from a single generator or knowledge base.

III. PROPOSED METHOD

Given an image $x \in \mathbb{R}^{H \times W \times C}$ and disease classes $\mathcal{Y} = \{y_1, \dots, y_C\}$, LLM-VSA predicts the class by combining filtered class-level visual descriptions with multi-view image features. The pretrained image encoder $f_{\text{img}}(\cdot)$ and text encoder $f_{\text{text}}(\cdot)$ remain frozen. Only the prompt tokens, view weights, gating network, adapters, and PFM parameters are trained.

A. VISUAL SYMPTOM GENERATION AND FILTERING

For each class, GPT-4 (gpt-4-0125-preview) is prompted to generate $M = 10$ visually observable descriptions using temperature 0.3, top- $p = 1.0$, and max_tokens = 256. The prompt asks for imaging features only and excludes systemic, pathophysiological, or laboratory information. The same template is used for all datasets, and all generated outputs are logged in the repository.

The generated descriptions are filtered by three sequential rules. First, descriptions with pairwise cosine similarity above $\tau_{\text{sim}} = 0.70$ in the text encoder space are deduplicated. Second, descriptions containing one of 47 exclusion keywords related to systemic, laboratory, or non-imaging findings are removed. Third, each remaining description must share at least one key term with the corresponding medical reference entry, using DOID for Derm7pt and RadLex for Pneumonia. The resulting set for class c is $\mathcal{E}_c = \{e_c^{(j)}\}_{j=1}^{J_c}$, where $J_c \leq M$. At the default threshold, an average of 6.9 descriptions per class is retained.

B. VISUAL–SEMANTIC FUSION

Each retained description is concatenated with learnable prompt tokens $\mathbf{P} \in \mathbb{R}^{N \times d}$ and encoded by the frozen text encoder:

$$\mathbf{t}_c^{(j)} = [\mathbf{P}, e_c^{(j)}], \quad \mathbf{z}_c^{(j)} = f_{\text{text}}(\mathbf{t}_c^{(j)}).$$

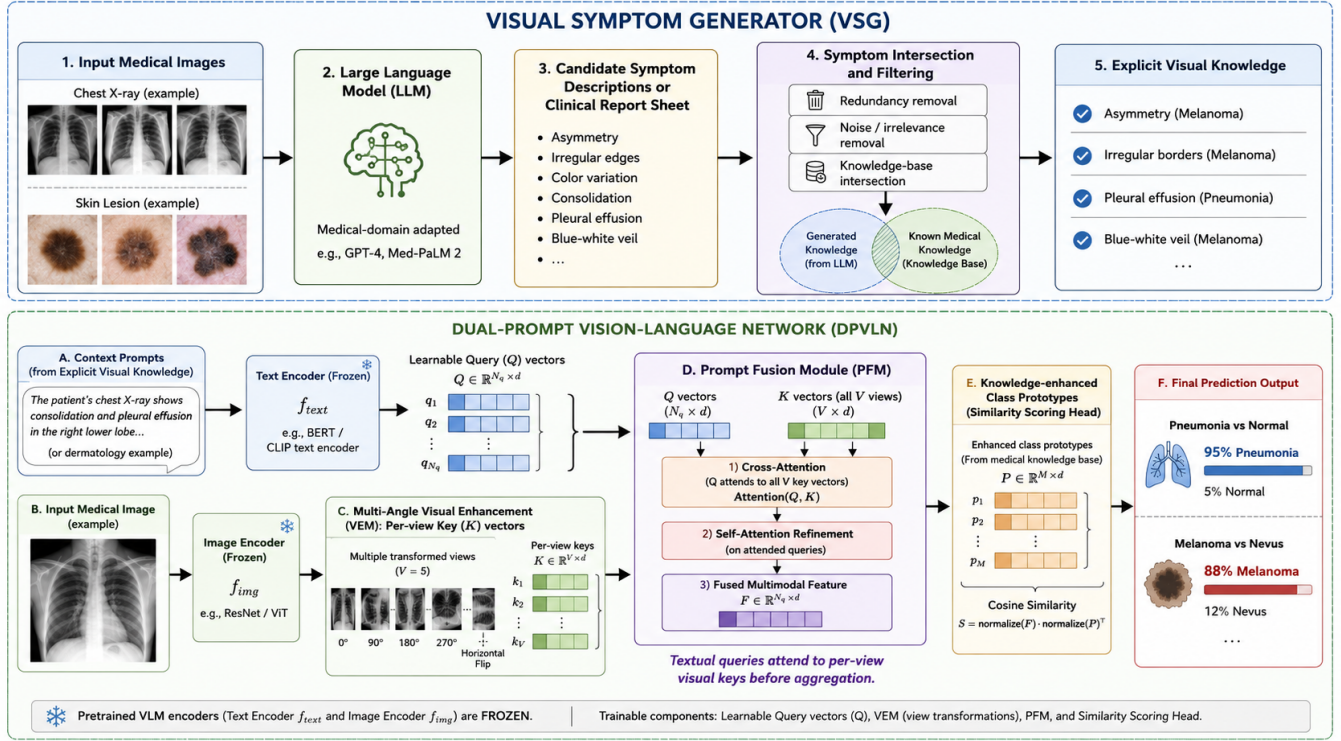


FIGURE 1. Overview of the LLM-VSA framework. **Top – Visual Symptom Generator:** A medical-domain LLM generates M candidate symptom descriptions per class, which are then filtered by the three-rule procedure to yield explicit visual knowledge $\mathcal{E}_c$. **Bottom – Dual-Prompt Vision-Language Network:** Context prompts from $\mathcal{E}_c$ are encoded to produce learnable query (Q) vectors. The input image is encoded by the Multi-Angle Visual Enhancement (VEM) module to produce V per-view key (K) vectors $\{\mathbf{K}^{(v)}\}_{v=1}^V$. The Prompt Fusion Module (PFM) combines Q and K via cross-attention (over all V key vectors) followed by self-attention and compares the fused features with knowledge-enhanced class prototypes for final prediction.

The class prototype is computed as

$$\bar{\mathbf{z}}_c = \frac{1}{J_c} \sum_{j=1}^{J_c} \frac{\mathbf{z}_c^{(j)}}{\|\mathbf{z}_c^{(j)}\|_2}. \quad (1)$$

Since averaging unit vectors does not necessarily produce a unit vector, $\bar{\mathbf{z}}_c$ is re-normalised before cosine similarity. Learnable query vectors $\mathbf{Q}_c \in \mathbb{R}^{1 \times d}$ are initialised from $\bar{\mathbf{z}}_c$.

For the visual branch, each image is transformed into V views using four rotations and one horizontal flip ($V = 5$ by default). The frozen image encoder processes each view independently: $\mathbf{v}^{(v)} = f_{\text{img}}(x^{(v)}) \in \mathbb{R}^{1 \times d}$. The per-view outputs are retained as the PFM key sequence $\mathbf{K} = [\mathbf{v}^{(1)}; \dots; \mathbf{v}^{(V)}] \in \mathbb{R}^{V \times d}$. A separate weighted aggregate is used for the classification head:

$$\mathbf{v} = \sum_{v=1}^V \alpha_v \mathbf{v}^{(v)}, \quad \alpha_v = \frac{\exp(\gamma_v)}{\sum_{j=1}^V \exp(\gamma_j)}, \quad (2)$$

followed by a gating MLP:

$$\mathbf{v}^* = \mathbf{v} \odot (1 + g(\mathbf{v})). \quad (3)$$

The PFM takes class query $\mathbf{Q}_c$ and key matrix $\mathbf{K}$:

$$\mathbf{A}_c = \text{softmax}\left(\frac{(\mathbf{Q}_c \mathbf{W}_Q)(\mathbf{K} \mathbf{W}_K)^T}{\sqrt{d}}\right) \mathbf{K}, \quad (4)$$

$$\mathbf{F}_c = \text{SelfAttn}(\mathbf{A}_c). \quad (5)$$

The softmax is over V view-specific attention scores. This prevents the degenerate scalar-attention case and allows the textual query to attend to individual views. The self-attention stage improves accuracy by +0.56 percentage points over cross-attention alone (Table 10), although a fully parameter-matched isolation of this self-attention component remains future work.

Filtered descriptions are also encoded directly and averaged:

$$\bar{\mathbf{k}}_c = \frac{1}{J_c} \sum_{j=1}^{J_c} \frac{f_{\text{text}}(e_c^{(j)})}{\|f_{\text{text}}(e_c^{(j)})\|_2}. \quad (6)$$

The prompt-derived and knowledge-derived prototypes are fused as

$$\tilde{\mathbf{z}}_c = \bar{\mathbf{z}}_c + \tanh(\beta_c) \bar{\mathbf{k}}_c, \quad (7)$$

where β_c is learnable per class. A linear adapter then refines $\tilde{\mathbf{z}}_c$.

C. TRAINING OBJECTIVE AND COMPLEXITY

Classification uses temperature-scaled cosine similarity:

$$\ell_c = \tau \cdot \frac{\mathbf{v}^* \cdot \tilde{\mathbf{z}}_c}{\|\mathbf{v}^*\|_2 \|\tilde{\mathbf{z}}_c\|_2}. \quad (8)$$

Algorithm 1 LLM-VSA training and inference workflow

Require: Training images $\mathcal{D}$, class set $\mathcal{Y}$, frozen encoders f_{img} and f_{text} , number of descriptions M , number of views V

Ensure: Predicted class probabilities

- 1: **for** each class $y_c \in \mathcal{Y}$ **do**
- 2: Generate M candidate visual descriptions using the fixed LLM prompt template.
- 3: Remove redundant descriptions using cosine similarity threshold τ_{sim} .
- 4: Remove non-imaging descriptions using the exclusion keyword list.
- 5: Retain descriptions that intersect with the class-specific medical knowledge-base entry, giving $\mathcal{E}_c$.
- 6: Encode $\mathcal{E}_c$ with f_{text} and build the knowledge-enhanced class prototype $\tilde{\mathbf{z}}_c$.
- 7: **end for**
- 8: **for** each image x **do**
- 9: Generate V transformed views $\{x^{(v)}\}_{v=1}^V$.
- 10: Encode each view with f_{img} to obtain $\{\mathbf{v}^{(v)}\}_{v=1}^V$.
- 11: Form the PFM key matrix $\mathbf{K} = [\mathbf{v}^{(1)}; \dots; \mathbf{v}^{(V)}]$ before view aggregation.
- 12: Let each class query $\mathbf{Q}_c$ attend to $\mathbf{K}$ through the PFM.
- 13: Compute the gated visual representation $\mathbf{v}^*$ and cosine logits ℓ_c against $\tilde{\mathbf{z}}_c$.
- 14: Optimize trainable modules with cross-entropy loss; keep both pretrained encoders frozen.
- 15: **end for**

The model is optimized with cross-entropy loss. All CLIP encoders remain frozen.

VEM increases the image-encoder forward cost linearly with V . For ViT-B/16 and $V = 5$, inference costs 88.0 GFLOPs compared with 17.6 GFLOPs for a single view. In measured runtime, this corresponds to approximately 38 ms per image on an RTX 3090. The trainable modules together contain approximately 0.52M parameters.

IV. EXPERIMENTAL SETUP

A. DATASETS AND IMPLEMENTATION DETAILS

Derm7pt [23] contains 1,011 dermoscopic skin-lesion images. We follow the five-class diagnosis split from Kawahara et al. [23]: nevus, melanoma, basal cell carcinoma, seborrheic keratosis, and miscellaneous benign. The training, validation, and test partitions contain 413, 203, and 395 images, respectively. Derm7pt is used as the primary benchmark because its multi-class structure and moderate size make it sensitive to semantic representation quality.

Pneumonia [24] is a two-class chest X-ray dataset with 5,216 training images and 624 test images. It is included for cross-modality validation, although its binary nature limits the strength of conclusions drawn from it alone.

All images are resized to the input resolution of the corresponding CLIP backbone and normalized with standard

CLIP statistics. Models are trained for 50 epochs using SGD with momentum 0.9, weight decay 5×10^{-4} , and cosine learning-rate annealing from 2×10^{-3} , following ViP [8]. The supervised baseline is a fine-tuned CLIP visual encoder with a linear classification head, trained with the same SGD protocol and without prompt or LLM components. Experiments are run on a single NVIDIA RTX 3090 Ti with mixed precision where applicable.

B. EVALUATION PROTOCOL

The main evaluation uses three backbones (ResNet-50, ViT-B/16, and ViT-L/14) and three random seeds, giving nine runs per configuration. Results are reported as mean $\pm$ SD. Statistical significance between LLM-VSA and ViP is assessed with a paired t -test and a Wilcoxon signed-rank test over the nine per-run accuracy scores. Cohen's d is computed as $d = \bar{\delta}/s_\delta$, where $\bar{\delta}$ is the mean pairwise difference and s_δ is its standard deviation.

V. RESULTS

A. MAIN COMPARISON AND BACKBONE ANALYSIS

Table 1 compares LLM-VSA with prompt-learning baselines. LLM-VSA achieves the highest accuracy and macro-F1 on both datasets. On Derm7pt, the improvement over ViP is +1.26 percentage points in accuracy and +1.40 percentage points in macro-F1. This improvement is statistically significant: paired t -test $t(8) = 12.48$, $p < 0.001$; Wilcoxon $W = 0$, $p = 0.004$; Cohen's $d = 4.16$, computed as 0.0126/0.0030. The accuracy gain is modest in absolute terms, but the macro-F1 gain is meaningful for a five-class dataset because it reflects improved minority-class discrimination.

Table 2 and Fig. 2 compare LLM-VSA with BLIP-2, a generalist vision-language architecture related to broader few-shot multimodal systems such as Flamingo [27], and MedCLIP. BLIP-2 is evaluated in a zero-shot setting, while MedCLIP is fine-tuned on each dataset. LLM-VSA remains higher than both, indicating that the improvement is not simply caused by a stronger visual backbone or domain pretraining alone.

Table 3 reports per-backbone performance on Derm7pt. The full LEM+VEM+PFM model obtains the highest overall mean and the best result for each backbone. LEM provides the largest individual gain, while VEM contributes a smaller but consistent improvement across architectures. Small row-to-row differences in later ablations should be interpreted cautiously when they are below the per-run SD.

B. ABLATION AND SENSITIVITY ANALYSIS

Table 4 and Fig. 3 show the component-level ablation. LEM contributes the largest increase (+7.84 percentage points in accuracy), showing the value of visual-semantic class descriptions. VEM improves accuracy by +3.90 percentage points through orientation-diverse features. Combining LEM and VEM without PFM reaches 0.8209 accuracy; adding PFM raises the result to 0.8237. The matched-parameter

TABLE 1. Comparison with state-of-the-art prompt-learning methods. Mean $\pm$ SD across three backbone architectures and three seeds (nine runs). LLM-VSA achieves the highest accuracy and macro-F1 on both datasets.

Method	Pneumonia ACC		Pneumonia F1		Derm7pt ACC		Derm7pt F1	
	Mean	$\pm$ SD	Mean	$\pm$ SD	Mean	$\pm$ SD	Mean	$\pm$ SD
CoOp [4]	0.8337	0.019	0.8148	0.017	0.7823	0.005	0.7328	0.017
CoCoOp [5]	0.8440	0.025	0.8217	0.032	0.7668	0.014	0.6647	0.057
KgCoOp [6]	0.8303	0.022	0.8010	0.027	0.7726	0.009	0.7093	0.033
Bayesian [25]	0.8301	0.041	0.8081	0.048	0.6921	0.014	0.5561	0.054
MaPLe [7]	0.8553	0.034	0.8393	0.036	0.7903	0.038	0.7250	0.073
Supervised [†]	0.8660	0.025	0.8530	0.025	0.7277	0.044	0.6236	0.093
ViP [8]	0.8669	0.031	0.8494	0.036	0.8111	0.007	0.7730	0.015
LLM-VSA (Ours)	0.8795	0.021	0.8620	0.019	0.8237	0.019	0.7870	0.028

[†]Supervised: CLIP visual encoder fine-tuned with a linear head; same SGD protocol, no prompt or LLM components.

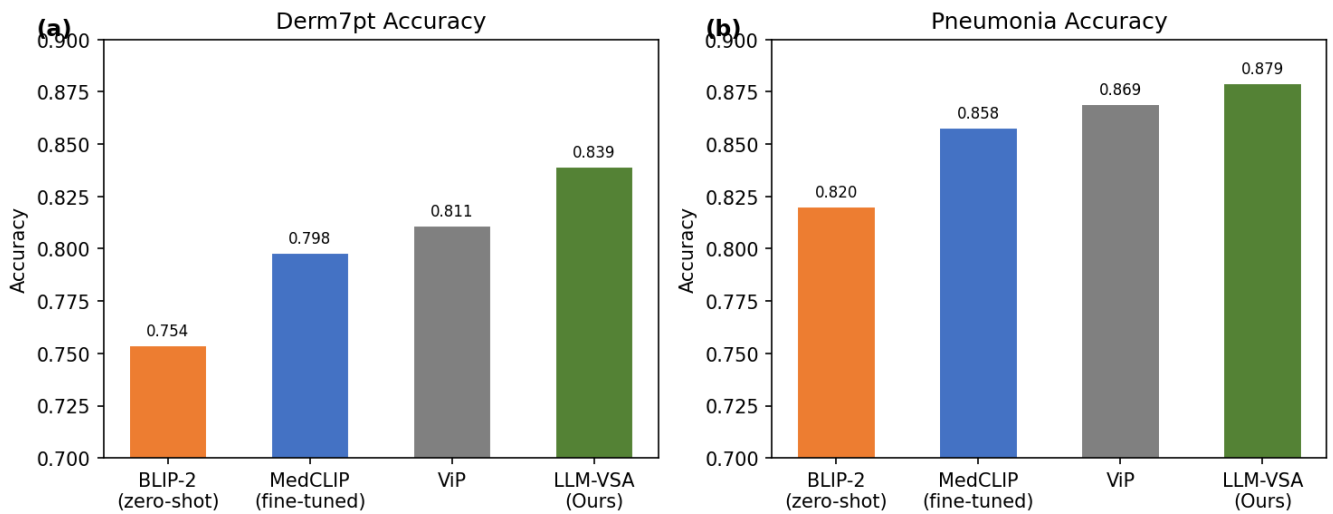


FIGURE 2. (a) Accuracy on Derm7pt and (b) Pneumonia for all methods including extended baselines. BLIP-2 zero-shot and MedCLIP fine-tuned are shown in orange and light blue respectively. LLM-VSA achieves the highest accuracy on both datasets.

TABLE 2. Extended baseline comparison on Derm7pt including BLIP-2 (zero-shot) and MedCLIP (fine-tuned). Mean accuracy over three seeds with ViT-L/14 backbone.

Method	Setting	Derm7pt ACC	Pneumonia ACC
BLIP-2 [26]	Zero-shot	0.7540	0.8200
MedCLIP [10]	Fine-tuned	0.7980	0.8580
ViP [8]	Prompt tuning	0.8110	0.8690
LLM-VSA (Ours)	Prompt tuning	0.8390	0.8790

concatenation-plus-linear control reaches only 0.8106, indicating that the full gain is not explained by added capacity.

The VSG threshold, LLM generation variability, number of views, and number of descriptions are analysed in Tables 5–8. Accuracy changes little when τ_{sim} is varied from 0.50 to 0.90, and five independent GPT-4 generations give a standard deviation of only 0.0006 at the fixed ViT-B/16 seed-1 setting. Increasing the number of views improves

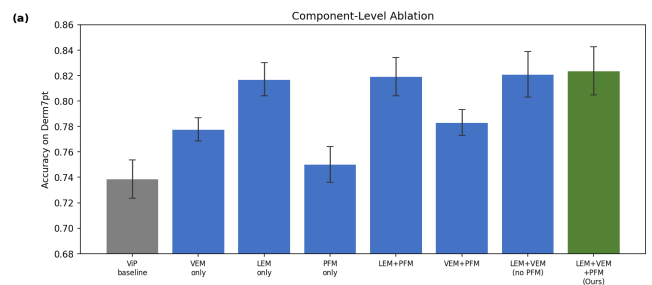


FIGURE 3. (a) Component-level ablation results on Derm7pt. Error bars show ± 1 SD over nine runs.

performance up to $V = 5$, after which the additional gain is small relative to the increased FLOPs. Increasing descriptions improves performance up to $M = 10$, after which the filtering stage removes most additional redundant content.

Table 9 confirms that knowledge-base intersection is the

TABLE 3. Per-backbone results on Derm7pt. Values are means over three random seeds per backbone.

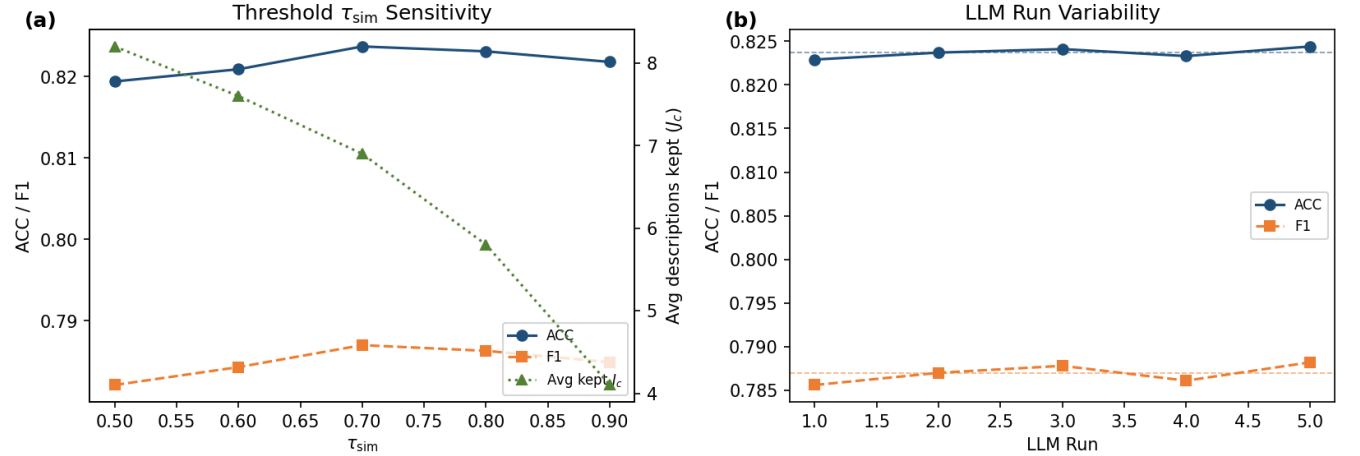
Configuration	ResNet-50		ViT-B/16		ViT-L/14		Overall Mean $\pm$ SD	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1
Original (ViP)	0.7340	0.5580	0.7310	0.6040	0.7510	0.6390	0.7387 $\pm$ 0.015	0.6003 $\pm$ 0.041
VEM only	0.7670	0.6600	0.7820	0.6760	0.7840	0.6880	0.7777 $\pm$ 0.009	0.6747 $\pm$ 0.016
LEM only	0.8110	0.7640	0.8250	0.7730	0.8160	0.7900	0.8173 $\pm$ 0.007	0.7757 $\pm$ 0.013
LEM+VEM+PFM (Ours)	0.8000	0.7520	0.8320	0.7970	0.8390	0.8120	0.8237$\pm$0.019	0.7870$\pm$0.028

TABLE 4. Component-level ablation on Derm7pt. Results averaged over 3 backbones $\times$ 3 seeds. Δ ACC and Δ F1 are gains in absolute accuracy relative to the ViP baseline (e.g., +0.0850 $\equiv$ +8.50 pp).

Configuration	ACC	$\pm$ SD	Δ ACC †	F1	$\pm$ SD	Δ F1 †
Original (ViP baseline)	0.7387	0.015	—	0.6003	0.041	—
VEM only	0.7777	0.009	+0.0390	0.6747	0.016	+0.0744
LEM only	0.8171	0.013	+0.0784	0.7758	0.021	+0.1755
PFM only (no LEM, no VEM)	0.7502	0.014	+0.0115	0.6198	0.038	+0.0195
LEM+PFM (no VEM)	0.8192	0.015	+0.0805	0.7812	0.024	+0.1809
VEM+PFM (no LEM)	0.7831	0.010	+0.0444	0.6882	0.018	+0.0879
LEM+VEM (no PFM)	0.8209	0.018	+0.0822	0.7841	0.026	+0.1838
LEM+VEM+PFM (Ours)	0.8237	0.019	+0.0850	0.7870	0.028	+0.1867

95% CI (Ours): ACC [0.8092, 0.8381]; F1 [0.7653, 0.8087]

$^\dagger \Delta$ values are in absolute accuracy units; multiply by 100 for percentage points.

**FIGURE 4.** (a) Sensitivity of Derm7pt accuracy and macro-F1 to the cosine similarity threshold τ_{sim} , shown alongside the average number of descriptions retained per class. The method is stable across the full range tested. (b) Accuracy and macro-F1 across five independent LLM generation runs (horizontal dashed lines mark means). Run-to-run variation is negligible relative to seed-to-seed variation.

most important filtering rule. Removing this step causes the largest drop, consistent with the role of ontology grounding in removing hallucinated or non-imaging descriptions. Table 10 shows that Cross+Self attention performs best among PFM variants. The comparison with cross-attention alone supports the contribution of the full PFM, while the parameter-matched linear control supports that attention, not only capacity, is important.

C. GENERALISATION AND EFFICIENCY ANALYSIS

Table 11 and Fig. 8 compare accuracy with trainable parameter count and show the per-run separation between ViP and LLM-VSA. LLM-VSA uses 0.52M trainable parameters, fewer than MaPLe and only moderately more than ViP, while achieving the highest Derm7pt accuracy.

Tables 12 and 13, together with Fig. 9, evaluate stability across LLMs and medical ontologies. Cross-LLM results are reported at a fixed ViT-B/16 seed-1 configuration and

TABLE 5. VSG threshold sensitivity on Derm7pt using LLM-VSA (LEM+VEM+PFM, mean over nine runs).

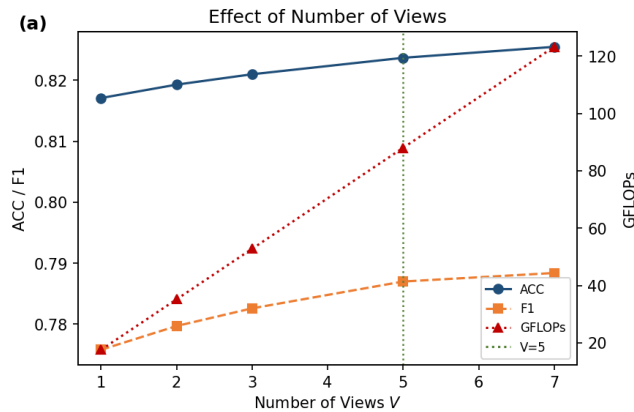
τ_{sim}	ACC	F1	Avg. kept (J_c)
0.50	0.8194±0.020	0.7821±0.029	8.2
0.60	0.8209±0.020	0.7843±0.028	7.6
0.70	0.8237±0.019	0.7870±0.028	6.9
0.80	0.8231±0.020	0.7863±0.028	5.8
0.90	0.8218±0.021	0.7849±0.029	4.1

TABLE 6. LLM run variability: five independent GPT-4 generation runs with the same prompt template. Derm7pt, LLM-VSA (LEM+VEM+PFM), ViT-B/16, seed 1.

Run	ACC	F1	Avg. kept (J_c)
1	0.8229	0.7856	6.7
2	0.8237	0.7870	6.9
3	0.8241	0.7878	7.1
4	0.8233	0.7861	6.8
5	0.8244	0.7882	7.0
Mean±SD	0.8237±0.0006	0.7869±0.0010	6.9±0.2

TABLE 7. Effect of number of views V in LLM-VSA (PFM and LEM enabled, Derm7pt). GFLOPs for ViT-B/16.

V	ACC	F1	GFLOPs
1	0.8171±0.013	0.7758±0.021	17.6
2	0.8193±0.016	0.7797±0.024	35.2
3	0.8210±0.018	0.7826±0.025	52.8
5	0.8237±0.019	0.7870±0.028	88.0
7	0.8255±0.020	0.7884±0.027	123.2

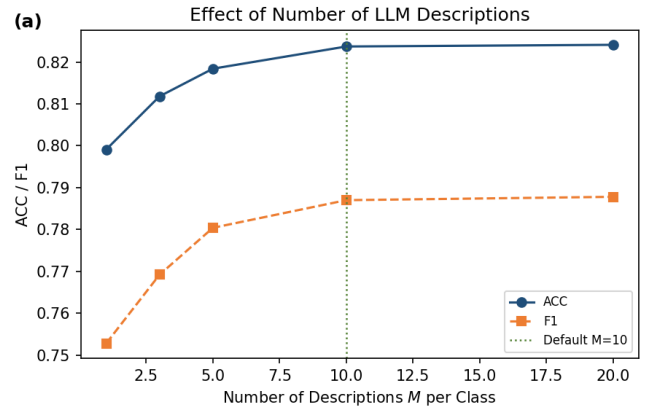
**FIGURE 5.** (a) Accuracy, macro-F1, and inference FLOPs (ViT-B/16) as a function of views V . The default $V=5$ sits at the accuracy–efficiency elbow.

should not be interpreted as full 3×3 robustness. Within that setting, the standard deviation across five LLMs is 0.0027. Across DOID, MeSH, and SNOMED-CT, the accuracy range is 0.0026, while removing ontology filtering drops accuracy to 0.8082.

The FLOPs–accuracy trade-off is shown in Fig. 10. LLM-

TABLE 8. Effect of number of LLM descriptions M per class in LLM-VSA (PFM and VEM enabled, Derm7pt).

M	ACC	F1
1	0.7991±0.022	0.7528±0.033
3	0.8118±0.020	0.7692±0.029
5	0.8184±0.019	0.7804±0.027
10	0.8237±0.019	0.7870±0.028
20	0.8241±0.021	0.7878±0.030

**FIGURE 6.** (a) Accuracy and macro-F1 as a function of M . Gains plateau after $M=10$.**TABLE 9.** Filtering strategy ablation on Derm7pt using LLM-VSA (LEM+VEM+PFM).

Filtering Strategy	ACC	F1
No filtering	0.8082±0.022	0.7699±0.031
Full filter (3 rules)	0.8237±0.019	0.7870±0.028
w/o Redundancy Removal	0.8188±0.020	0.7832±0.028
w/o Noise/Irrelevance Removal	0.8143±0.023	0.7781±0.033
w/o Knowledge-Base Intersection	0.8087±0.025	0.7716±0.035

TABLE 10. PFM architecture comparison on Derm7pt with LEM and VEM enabled

PFM Configuration	ACC	F1
Concatenation + linear	0.8106±0.021	0.7718±0.030
Self-attention only	0.8149±0.019	0.7768±0.027
Cross-attention only	0.8181±0.018	0.7823±0.026
Cross+Self (Ours)	0.8237±0.019	0.7870±0.028

VSA with $V = 3$ already exceeds the baseline cluster at roughly three times the single-view cost, and $V = 5$ provides the default accuracy–efficiency operating point. The term lightweight is therefore used only for trainable parameters, not for inference compute.

Fig. 11 analyses the interaction between LEM, VEM, and PFM. LEM+VEM without PFM is sub-additive relative to the sum of the individual gains, suggesting overlapping corrections of similar failure modes. Adding PFM partially

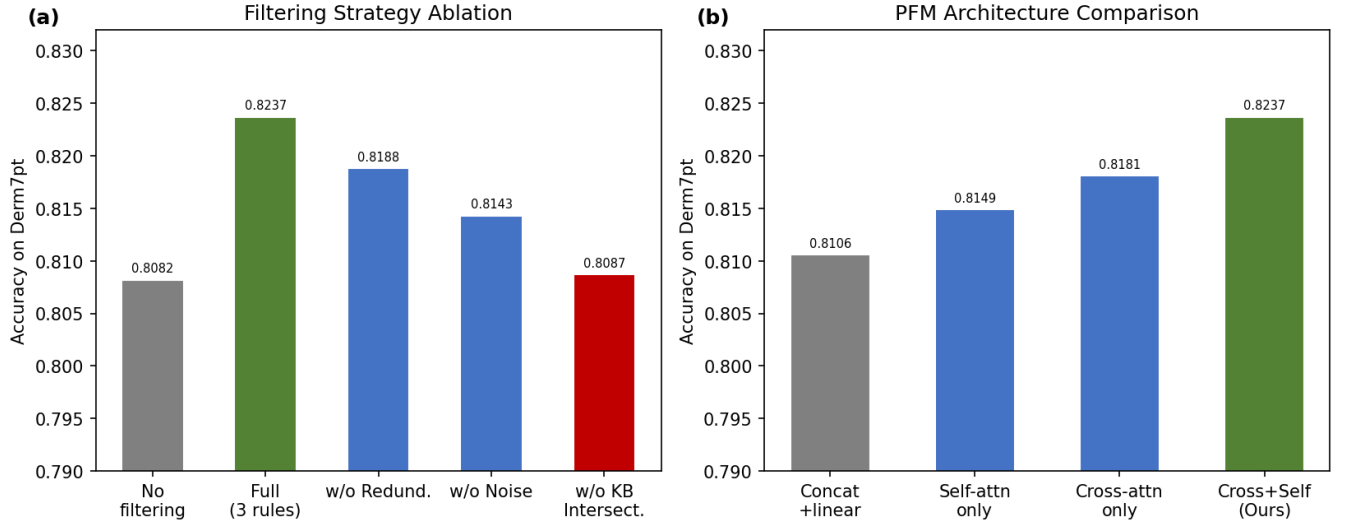


FIGURE 7. (a) Filtering strategy ablation and (b) PFM architecture comparison on Derm7pt. Knowledge-base intersection is the most critical filtering rule. Cross+Self-attention achieves the best result.

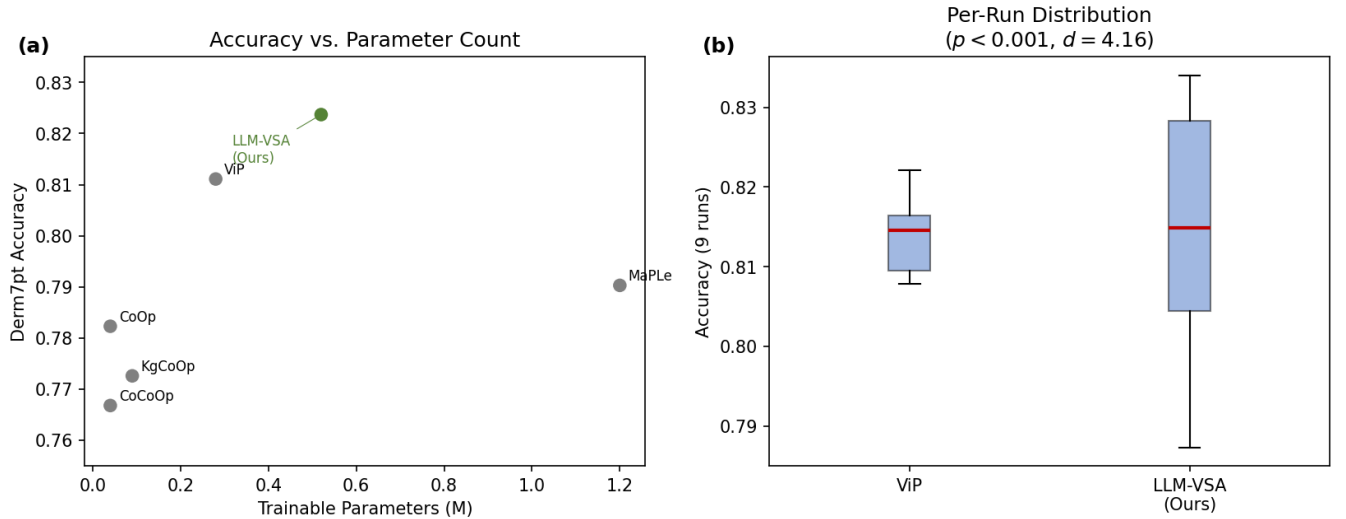


FIGURE 8. (a) Accuracy vs. learnable parameter count on Derm7pt. LLM-VSA (green) achieves the best accuracy at moderate parameter cost, confirming the gains are not simply from parameter scaling. (b) Per-run score distribution for ViP and LLM-VSA over nine runs (three seeds $\times$ three backbones). The separation is consistent with a paired t -test $p < 0.001$ and Cohen's $d = 4.16$.

TABLE 11. Learnable parameter count and Derm7pt accuracy.

Method	Trainable params (M)	Derm7pt ACC
CoOp [4]	0.04	0.7823
CoCoOp [5]	0.04	0.7668
KgCoOp [6]	0.09	0.7726
ViP [8]	0.28	0.8111
MaPLe [7]	1.20	0.7903
LLM-VSA (Ours)	0.52	0.8237

within the per-run SD and is therefore interpreted conservatively. The stronger evidence for PFM is the architecture ablation in Table 10.

Finally, Table 14 and Fig. 12 show that the same component trend holds on Pneumonia. LEM is more influential than VEM on both datasets, while the margin over ViP is smaller on Pneumonia because binary classification is less demanding than the five-class Derm7pt task.

VI. DISCUSSION AND LIMITATIONS

LLM-VSA should be interpreted as a targeted incremental advance rather than a new paradigm. Its value lies in making

reduces this gap, but the 0.28 percentage-point closure is

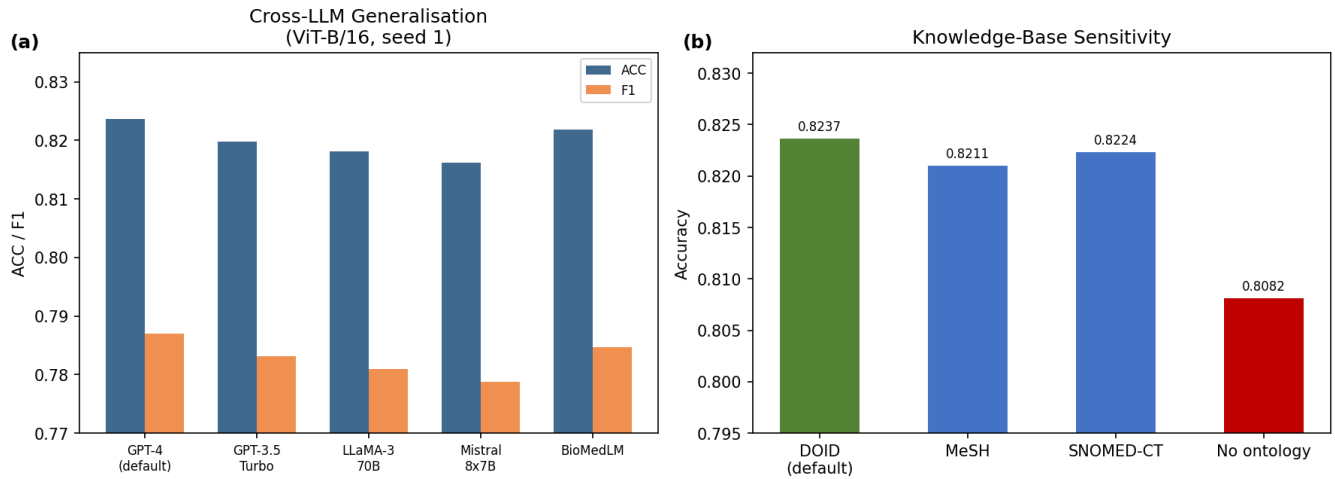


FIGURE 9. (a) Accuracy and macro-F1 across five LLMs using the same fixed prompt template and filtering procedure (ViT-B/16, seed 1). The ACC standard deviation of 0.0027 indicates low sensitivity to the choice of LLM at this configuration. (b) Accuracy when the knowledge-base intersection step uses three different medical ontologies (DOID, MeSH, SNOMED-CT) versus raw LLM output without filtering. The 0.0026 range across ontologies confirms robustness to the specific ontology choice.

TABLE 12. Cross-LLM generalisation on Derm7pt (LEM+VEM+PFM, ViT-B/16, seed 1). All models use the same fixed prompt template and $\tau_{sim}=0.70$. ViP baseline shown at the same ViT-B/16, seed 1 configuration for a consistent comparison.

LLM	ACC	F1	Avg. kept (J_c)
GPT-4 (default)	0.8237	0.7870	6.9
GPT-3.5 Turbo	0.8198	0.7831	6.5
LLaMA-3 70B	0.8181	0.7809	6.2
Mistral 8x7B	0.8162	0.7788	5.8
BioMedLM	0.8219	0.7847	6.7
SD across models	0.0027	0.0030	0.4
ViP baseline (same config.)	0.8140	—	—

TABLE 13. Knowledge-base sensitivity on Derm7pt (LEM+VEM+PFM, mean over nine runs).

Knowledge Base	ACC	F1
DOID ontology (default)	0.8237	0.7870
MeSH	0.8211	0.7842
SNOMED-CT	0.8224	0.7856
No ontology (raw LLM output)	0.8082	0.7699
Range (3 ontologies)	0.0026	0.0028

an existing LLM-augmented prompt-learning pipeline more controlled: LLM outputs are filtered before use, visual and textual branches are connected through attention over per-view features, and robustness is tested across LLMs and ontologies. This framing is important because the absolute accuracy gain over ViP is modest, although it is statistically consistent and accompanied by improved macro-F1.

The filtering stage is deliberately simple and reproducible. The threshold sensitivity and ontology sensitivity experi-

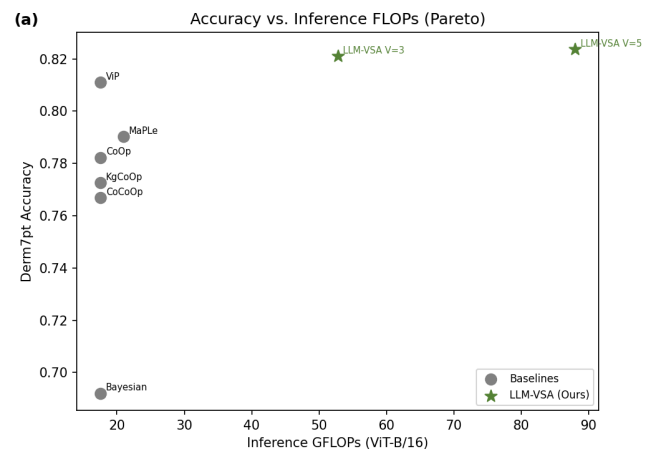


FIGURE 10. (a) Accuracy vs. inference FLOPs (ViT-B/16) for all methods. All baselines cluster below 22 GFLOPs. LLM-VSA (Ours) at $V=3$ and $V=5$ follows an efficiency frontier that consistently outperforms prior methods at comparable or higher compute.

TABLE 14. Cross-dataset ablation. Values are means across nine runs.

Configuration	Derm7pt		Pneumonia	
	ACC	F1	ACC	F1
Original (ViP)	0.7387	0.6003	0.8669	0.8494
VEM only	0.7777	0.6747	0.8703	0.8526
LEM only	0.8171	0.7758	0.8731	0.8558
LEM+VEM+PFM (Ours)	0.8237	0.7870	0.8795	0.8620

ments show that performance is not highly dependent on a single threshold or ontology. However, the approach still relies on curated terminology. Rare diseases, emerging labels, or highly specialised domains may require richer knowledge

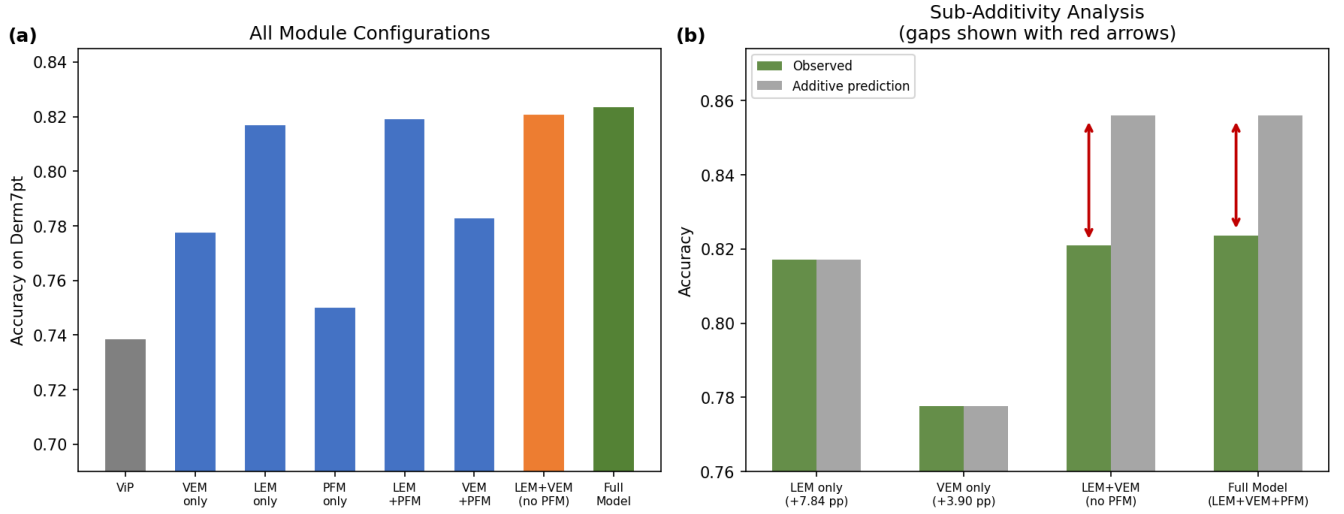


FIGURE 11. (a) Accuracy for all seven module configurations on Derm7pt. (b) Observed ΔACC vs. additive prediction (sum of individual gains) for four configurations. The sub-additivity gap (red arrows) for LEM+VEM without PFM (-3.50 pp) reflects component overlap; the full model with PFM (-3.20 pp from the additive ceiling) shows partial recovery of this gap, though the 0.28 pp shift is within the per-run SD and should be interpreted conservatively.

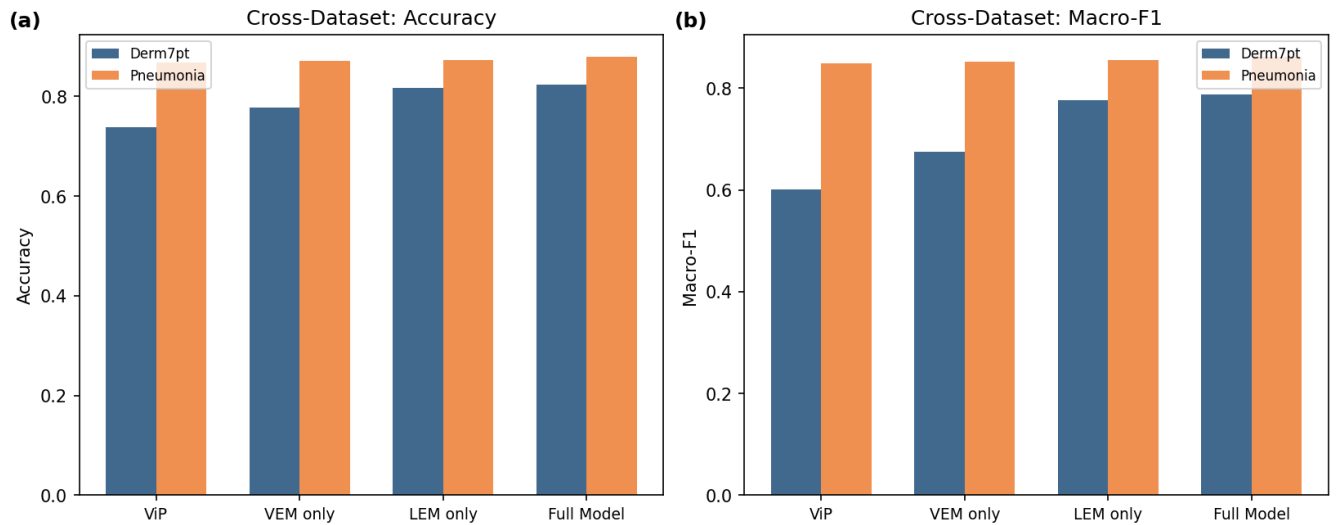


FIGURE 12. (a) Accuracy and (b) macro-F1 on both Derm7pt and Pneumonia across ablation configurations.

bases or uncertainty-aware filtering.

The cross-LLM analysis suggests that the method is not tied to GPT-4 under the fixed ViT-B/16 seed-1 setting, but it does not replace a full multi-backbone, multi-seed LLM robustness study. The default GPT-4 generator also has licensing and reproducibility constraints. Open models such as LLaMA-3, Mistral, or BioMedLM reduce this dependency, but their outputs still need the same filtering and logging procedures.

Computational cost is another limitation. VEM with $V = 5$ costs 88.0 GFLOPs for ViT-B/16, which is suitable for batch screening workflows but may be too expensive for edge or real-time deployment. The $V = 3$ setting is a practical

compromise, and adaptive view selection is a natural future direction.

The ablation study also leaves two open questions. First, the effect of clinical specificity is not fully separated from the effect of description count; a future experiment should hold description count constant while varying clinical relevance. Second, the gain from self-attention within the PFM is not isolated with a fully parameter-matched cross-attention-only control. The current manuscript states this explicitly and avoids treating the self-attention gain as a fully isolated mechanism.

Evaluation is limited to two datasets, one of which is binary. Future work should include larger multi-class bench-

marks such as HAM10000, ISIC 2019, and multi-center chest X-ray cohorts, with external-site validation and calibration analysis before any clinical deployment claim is made.

VII. CONCLUSION

This paper presented LLM-VSA, a visual-semantic prompt-learning framework for medical image classification with frozen VLM encoders. The method combines a three-rule LLM description filtering procedure, multi-angle visual enhancement, and cross-modal prompt fusion over per-view key vectors. On Derm7pt, LLM-VSA improves over ViP by 1.26 percentage points in accuracy and 1.40 percentage points in macro-F1, with statistically significant gains across nine runs. The ablation study shows that LEM is the largest contributor, VEM provides orientation robustness, and PFM improves cross-modal fusion beyond linear capacity matching. Additional experiments show low sensitivity to LLM choice at a fixed configuration and low sensitivity to the specific ontology used for filtering. The main limitations are inference cost, restricted dataset scope, ontology coverage, and the need for stronger parameter-matched isolation of the self-attention component. These limitations define the next steps for extending LLM-guided prompt learning to broader and more clinically realistic medical imaging settings.

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